Sourabh Singh

sourabhsingh2003ltp@gmail.com | +91-9695098754

AI Engineer specializing in Python, Machine Learning, Generative AI, LLMs, RAG, and AI agent development. Skilled in building scalable AI applications, integrating LLMs, and developing intelligent automation solutions using modern AI frameworks and APIs..

Education

B.Tech in Computer Science Engineering *2022 – 2026*
Ajay Kumar Garg Engineering College, Ghaziabad, Uttar Pradesh

Technical Skills

Languages:	Python, SQL, JavaScript, TypeScript
AI / Machine Learning:	Machine Learning, Deep Learning, NLP, Generative AI, Scikit-learn, TensorFlow, PyTorch
GenAI / LLMs:	OpenAI API, LLMs, LangChain, LangGraph, Prompt Engineering, RAG, AI Agents, Hugging Face
Vector Databases:	FAISS, Pinecone, ChromaDB
Frameworks / APIs:	FastAPI, Django, REST APIs, React.js, Next.js
Databases:	SQL, MongoDB
Tools / Deployment:	Git, GitHub, Docker, Linux/Bash, Postman, VS Code, AWS, Vercel
CS Fundamentals:	Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

Experience

AI engineer | *Convra* *Dec 2025- Present*

- Developed AI-powered applications using **Python, Generative AI, and Large Language Models (LLMs)**, integrating intelligent automation and AI-driven features into web applications.
- Implemented **Retrieval-Augmented Generation (RAG), LangChain, prompt engineering, and vector databases** to enable contextual question answering, document retrieval, and AI-powered search.
- Integrated **OpenAI APIs and AI agent workflows**, developed REST APIs, and optimized application performance to deliver scalable and reliable AI solutions.

AI/ML Intern | *Prodigy InfoTech* *2025*

- Developed and evaluated machine learning models using Python, Pandas, NumPy, and Scikit-learn for predictive analytics tasks.
- Performed data preprocessing, feature engineering, and exploratory data analysis (EDA) to identify patterns and improve model performance.
- Applied model evaluation techniques and created visualizations using Matplotlib and Seaborn to communicate findings and analytical insights.

Projects

MCP-Powered AI Assistant *2026*

- Developed a full-stack **AI assistant** with a **Next.js** chat interface and **FastAPI** backend, implementing **JWT authentication** and persistent, user-specific conversations.
- Integrated a local **Model Context Protocol (MCP)** server using the official **Python SDK** and **stdio transport** for dynamic tool discovery and execution.
- Implemented secure document uploads, **PDF/DOCX text extraction**, and **user-scoped keyword search** for document retrieval.
- Built a configurable **OpenAI-compatible LLM adapter** with environment-based secret management and **Docker Compose** for local development.

StockAlphas – AI-Powered Stock Analysis *2026*

- Built a full-stack AI-powered stock analysis platform with authentication, real-time watchlists, portfolio tracking, and Vercel deployment.
- Developed **10+ REST APIs** integrating Yahoo Finance, technical indicators (RSI, MACD, Moving Averages), and a Python ML microservice generating Buy/Sell/Hold signals.
- Integrated AI-powered insights and predictive analytics using Scikit-learn for real-time stock recommendations.
- **Skills:** Next.js, TypeScript, Node.js, MongoDB, Python, Scikit-learn, REST APIs, NextAuth.js, AI/ML.

Orynta – AI-Powered Web Agency Platform *2026*

- Built and deployed a production-grade AI-powered web platform using **Next.js App Router**, TypeScript, and Tailwind CSS.
- Integrated **OpenAI API**, LangChain, AI agents, and RAG workflows to automate lead capture, customer support, and content generation.
- Designed scalable, reusable UI components and optimized SEO, performance, and deployment pipelines using Vercel and GitHub.
- **Skills:** Next.js, TypeScript, Tailwind CSS, OpenAI API, LangChain, RAG, AI Agents, Vercel, GitHub.

Certifications

- **IBM SkillsBuild – Agentic AI: From Learner to Builder**
- **Python & Django: Backend Web Development (Udemy)**
- **React.js Frontend Development (Udemy)**

Achievements & Activities

- Selected participant in **Viksit Uttar Pradesh @2047** — Resolution to Prosperity initiative.
- **Winner, IDE Bootcamp 2025 (JIIT)** for an innovative software solution.